D07 · Warehouse Operations & WMS

01 · Warehouse layout design & slotting optimization

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Layout design arranges receiving, storage, picking, packing and shipping zones, aisles and equipment to make flows short and uncongested; slotting assigns each SKU to the location that minimizes picking travel within its size, weight and compatibility constraints.

2 WHY IT MATTERS

Order picking is typically the most labor-intensive and costly warehouse activity, and travel is the largest share of picker time **(de Koster, Le-Duc and Roodbergen, 2007 — European Journal of Operational Research)**. **Because a small share of SKUs drives most of the picks, placing them well is one of the cheapest productivity gains available.** But demand shifts, so a slotting that was optimal a year ago decays quietly.

3 HOW IT IS DONE

1. **Analyze velocity and cube.** Picks per SKU, units per pick and dimensions over a representative period.
2. **Place fast movers in the golden zone.** Near dispatch, at waist height, with ergonomics and weight rules.
3. **Apply compatibility constraints.** Temperature, hazard class and food vs. non-food are hard rules, not preferences.
4. **Re-slot on a schedule.** At least twice a year and after major portfolio changes; track travel per pick.

IN PRACTICE

Frequency-based slotting delivers the fastest ROI of any warehouse initiative

In most distribution centers a small share of SKUs (often 10–20%) generates the majority of picking movements. Moving those SKUs to the shortest-travel zone has an immediate productivity impact.

Semi-annual re-slotting is the process that keeps the optimal slotting current

The SKU portfolio changes continuously — NPIs entering, discontinued products, seasonal shifts. The optimal slotting of 12 months ago is no longer optimal if the portfolio has changed significantly.

Warehouse layout must be designed before selecting equipment

The equipment type (counterbalance, narrow-aisle, VNA) determines the minimum aisle width — and aisle width determines available storage density. This sequence cannot be reversed without costly consequences.

Include compatibility restrictions in slotting — not just frequency

A slotting that only optimizes by picking frequency may place a food product and a chemical product together. Compatibility constraints (temperature, flammability, food vs. non-food) are hard constraints the slotting must respect.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Warehouse re-slotting — distribution DC, 8,400 active SKUs, 280 picking operators — Analysis reveals the top 200 highest-velocity SKUs are distributed throughout the warehouse with no frequency-based grouping.

Slotting action	SKUs affected	Productivity impact
Relocation of top 200 SKUs to golden zone (front, mid-level)	200	Picking travel for top SKUs: -62% · These SKUs represent 68% of all movements
Consolidation of slow-moving SKUs into deep storage zone	1,200	Space freed in mid zone: +18% for A-items · Space-neutral DIO impact

Slotting action	SKUs affected	Productivity impact
Bulky SKUs assigned to bottom level · Light SKUs to top level	3,400	Ergonomics improved · Handling incidents: -28%

Result: Picking productivity: 68 lines/hr -> 89 lines/hr (+31%). Walking distance/operator/day: 14.2 km -> 9.1 km (-36%). Re-slotting took 3 days to implement and generated $380K in annual picking cost savings.

HOW IT IS MEASURED

Lines Picked per Hour (LPPH) — picking productivity

Lines Picked per Hour (LPPH)Total picking lines completed / Total person-hours dedicated to picking

Benchmark: 60–100 LPPH for manual picking · 150–250 for assisted picking (voice, RF) · 200–400 for GTP An LPPH drop without a volume change indicates the slotting has deteriorated (high-velocity SKUs have migrated to suboptimal locations) or aisle congestion has increased.

Average Travel Distance per Pick

Average Travel Distance per PickTotal kilometers walked by picking operators / Total picks for the period

Benchmark: Continuous 5–15%/year reduction in active re-slotting programs · Typical baseline: 8–15 km/operator/day Travel distance per pick is the most direct slotting indicator. An increase without volume growth signals the optimal slotting is degrading as the active SKU portfolio changes.

WHAT YOU WOULD USE

Warehouse Planning

Manhattan Associates WMS — Slotting — Module: Slotting Optimization + Labor ManagementManhattan Associates' slotting module automatically calculates optimal locations for each SKU based on picking frequency, size, and compatibility restrictions.

SAP EWM — Warehouse Management — Module: Layout Design + Slotting RulesSAP Extended Warehouse Management with location management and configurable slotting rules by zone, SKU profile, and storage restrictions.

Analytics

Python (scipy / numpy) — Module: Slotting Optimization ModelPython-based slotting optimization models that calculate the optimal SKU-to-location assignment minimizing total picking travel distance.

4 SO WHAT

Put the right SKU in the right place, and keep it there. Slot by velocity, respect compatibility, re-slot twice a year and use travel per pick as the alarm.

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